MIPS ADC Change Detection Capability

The MIPS system supports detection of an analog voltage change on one of its ADC inputs, channels 0 through 3. A detected change in the analog voltage can be used to adjust one or more DC bias outputs, or the change can be used to trigger a timetable as well as adjust a time parameter in a timetable.

The ADC is a unipolar 0 to 5 volt, 12-bit input. A programmable gain of 1, 2, or 4 can also be defined by the user.

The ADC change detection allows you to define the ADC channel to use (0 through 3) as well as a window in ADC counts for change detection. The window is centered on the current ADC value. If a new ADC value falls outside the current ADC value by more than the window value, the change is detected and processed as configured by the user.

There are two filter options to minimize false detections. To detect a change, the ADC value must fall outside the window for a user-defined number of consecutive readings; this helps reject noise on the input signal. Additionally, the new value must remain within the new window for a user-defined number of readings before the change is reported, allowing the user to ignore transient fluctuations such as those seen on a filtered step change.

The ADC change detection system supports three operating modes; only one mode may be active at a time:

Change reporting	Report the detected change value on the MIPS host interface for tuning and diagnostics.
DC bias adjustment	Automatically adjust one or more DC bias channel voltages or the DC bias offset.
Timetable triggering	Trigger a timetable and optionally adjust a time point value in the timetable.

The following sections describe each mode and its associated commands. All commands are case-sensitive.

Initialization Commands

The following commands initialize and configure the ADC change detection system and must be issued before enabling any of the three operating modes.

ADCCHG,*channel*,*window*

Initializes the ADC change detection system. The *channel* parameter selects the ADC input to monitor (0 through 3). The *window* parameter defines the detection threshold in ADC counts, centered on the current ADC value. For example, a *window* of 10

signals a change when the ADC value deviates by more than ±10 counts from its current value.

ADCGAIN,*channel*,*gain*
Applies a programmable *gain* to the specified ADC *channel*. Valid *gain* values are 1, 2, or 4. This command must be issued after ADCCHG; if issued before, it will be ignored.

ADCCPRM,*wcnt*,*rcnt*
Adjusts the filter parameters to reduce false change detection, which is especially useful for noisy signals. *wcnt* defines the number of consecutive ADC readings that must fall outside the window before a change is detected; the default is 4. *rcnt* defines the number of readings that must remain within the new window before the change is reported; the default is 6.

ADCABORT
Stops the ADC change detection system and disables whichever operating mode is currently active.

Table 1, Initialization Commands

Command	Parameters	Description
ADCCHG	<channel>,<window>	Initializes the ADC change detection system. channel: 0–3. window: detection threshold in ADC counts (± centered on current value).
ADCGAIN	<channel>,<gain>	Sets the ADC input gain for the specified channel. Valid gain values: 1, 2, or 4. Must be issued after ADCCHG.
ADCCPRM	<wcnt>,<rcnt>	Sets filter parameters. wcnt: consecutive out-of-window readings required to signal a change (default 4). rcnt: readings within new window required before reporting (default 6).
RADCCHG	<gain>	Enables change-reporting mode. Reports the detected ADC change on the MIPS host interface. The gain parameter scales the ADC reading before it is reported.
ADCABORT		Stops the ADC change detection system and disables the active operating mode.

DC Bias Channel Voltage Adjustment

The detected ADC change can be applied to any number of DC bias channels or to the DC bias channel offset. Applying the change to the offset affects all output channels simultaneously. This feature is useful when the voltages in MIPS need to track an external reference signal.

All commands in this section include a *board* parameter that selects the DC bias module. The *board* value ranges from 0 to the number of installed DC bias modules minus 1. If only one module is present, its board address (0 or 1) depends on its jumper setting. Issue the ABOUT command to list installed modules and their addresses; address A corresponds to board 0 and address B corresponds to board 1.

SDCBOFOF,*board*,*value*
Sets the DC bias offset voltage to apply to the offset channel on the specified *board*. This is the same parameter adjusted automatically by the ADC change system. The *value* parameter is the offset voltage in volts.

GDCBOFOF,*board*
Returns the current DC bias offset voltage applied to the offset channel of the specified board.

SDCBCHOF,*board*,*value*
Sets the offset voltage to apply to one or more DC bias channels on the specified *board*. This is the same parameter adjusted automatically by the ADC change system. The *value* parameter is the offset voltage in volts. The channels affected are selected by SDCBCHMK.

GDCBCHOF,*board*
Returns the current offset voltage applied to the selected DC bias channels of the specified board.

SDCBCHMK,*board*,*mask*
Defines a hex bit mask that selects which DC bias channels are adjusted by the SDCBCHOF setting. The LSB of *mask* corresponds to channel 1 and the MSB to channel 8. For example, a *mask* of 01 enables only channel 1; a *mask* of FF enables all 8 channels.

GDCBCHMK,*board*
Returns the current bit mask defining the DC bias channels adjusted by SDCBCHOF.

DCBCADCOF,*board*,*gain*
Enables automatic adjustment of the DC bias offset value on the specified *board* in response to a detected ADC change. The detected ADC value is multiplied by *gain* and added to the board's DC bias offset. This command may be issued for multiple boards to adjust the bias offset on more than one board simultaneously.

DCBCADCCO,*board*,*gain*
Enables automatic adjustment of the selected DC bias channels on the specified *board* in response to a detected ADC change. The detected ADC value is multiplied by *gain* and added to the board's DC bias channel values. This command may be issued for multiple boards to adjust channel values across more than one board simultaneously.

SDCBCADCP,*board*,*DIO*
Assigns a digital input to control the polarity of the ADC adjustment for the specified *board*. When *DIO* is set to Q through X, the state of that digital input is read when an ADC change is detected. If the *DIO* input is logic high, the calculated voltage adjustment is multiplied by −1 to invert the polarity.

GDCBCADCP,*board*

Returns the DIO channel assigned for polarity detection on the specified board.

Table 2, DC Bias Voltage Adjustment Commands

Command	Parameters	Description
SDCBOFOF	<board>,<value>	Sets the DC bias offset voltage applied to the offset channel of the specified board.
GDCBOFOF	<board>	Returns the current DC bias offset voltage applied to the offset channel.
SDCBCHOF	<board>,<value>	Sets the offset voltage applied to the channels selected by SDCBCHMK.
GDCBCHOF	<board>	Returns the current offset voltage applied to the selected DC bias channels.
SDCBCHMK	<board>,<mask>	Hex bit mask selecting DC bias channels adjusted by SDCBCHOF. LSB = ch 1, MSB = ch 8. Example: 01 = ch 1 only, FF = all 8 channels.
GDCBCHMK	<board>	Returns the bit mask defining the DC bias channels adjusted by SDCBCHOF.
DCBCADCOF	<board>,<gain>	Enables automatic ADC-driven adjustment of the DC bias offset. ADC value × gain is added to the board's offset. May be issued for multiple boards.
DCBCADCCO	<board>,<gain>	Enables automatic ADC-driven adjustment of selected DC bias channels. ADC value × gain is added to the board's channel values. May be issued for multiple boards.
SDCBCADCP	<board>,<DIO>	Assigns a DIO input (Q–X) to control polarity. When the DIO is logic high at the time of detection, the voltage adjustment is multiplied by −1.
GDCBCADCP	<board>	Returns the DIO channel assigned for polarity detection on the specified board.

Timetable Triggering

The detected ADC change can be used to trigger a timetable and optionally adjust a time point count value within that timetable. Additional features allow conversion of the raw ADC value into engineering units using a user-defined gain and offset. A lookup table of up to 5 entries can then map the engineering unit value to a time in milliseconds. This chain of operations allows a detected voltage change to be automatically converted to an appropriate time count and applied to a running timetable.

The following commands configure and enable timetable triggering:

STPADJ,*count*,*channel*

Identifies a time point in the currently loaded timetable that will be adjusted by the ADC change system. The time point is specified by its current time *count* and one of the *channel* entries listed at that time point.

SADJRNG,*minimum*,*maximum*

Defines the allowed range for the adjusted time point count value. The adjusted time point cannot be set below the preceding time point or above the following time point

in the table. The user must define appropriate *minimum* and *maximum* values to stay within these constraints.

SADCMZCAL,*gain*,*offset*

Converts the raw ADC reading into engineering units (for example, m/z). The raw ADC value is multiplied by *gain* and then *offset* is added to produce the engineering unit value.

SMZTARG,*index*,*mz*,*time*

Populates an entry in the lookup table that maps engineering unit values to timetable count times in milliseconds. Up to 5 entries (*index* 0 through 4) can be defined. *mz* is the engineering unit value and *time* is the corresponding count time in milliseconds.

TRGTBLADC

Enables timetable triggering on ADC value change. When a change is detected, the ADC value is converted to engineering units via SADCMZCAL, looked up in the SMZTARG table, and the resulting time in milliseconds is converted to counts and used to adjust the selected time point in the timetable automatically.

Table 3, Timetable Triggering Commands

Command	Parameters	Description
STPADJ	<count>,<channel>	Identifies the timetable time point to adjust, specified by its current count value and one of its channels.
SADJRNG	<minimum>,<maximum>	Defines the allowed range for the adjusted time point count value. Must be set to keep the time point between its neighbors.
SADCMZCAL	<gain>,<offset>	Converts the raw ADC value to engineering units: (ADC value × gain) + offset.
SMZTARG	<index>,<mz>,<time>	Defines a lookup table entry (index 0–4) mapping an engineering unit value (mz) to a timetable count time in milliseconds (time).
TRGTBLADC		Enables timetable triggering on ADC change. Converts ADC value to engineering units, performs lookup table conversion, and adjusts the configured timetable time point automatically.